Digital Evidence Management System

DEMS | Secure tracking of digital evidence and chain of custody

Platform: Linux (Kali) | MariaDB + Bash | Presented by: Sameera

Problem and Objective

The Problem

Digital evidence (phone data, CCTV, emails, images) is easy to copy, alter or lose.
Paper registers and spreadsheets give no reliable history.
A gap in the handling record can make evidence inadmissible in court.

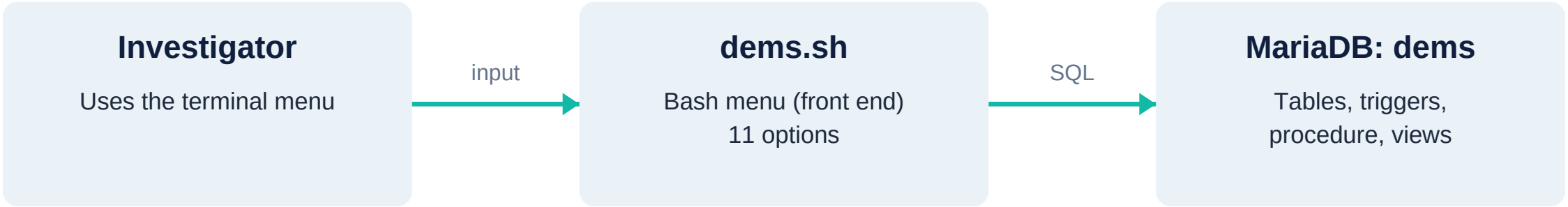
The Objective

Keep all cases and evidence in one central database.
Record every transfer: who, to whom, when, why.
Make the history tamper-proof and quick to search.

Result

A structured system that proves what the evidence is, where it is, and who handled it, at every step of the investigation.

System Architecture



Design principle: the script is only a front end. All rules and protection live inside the database, so a web or Python interface added later gets the same safety automatically.

Technology stack

Kali Linux | MariaDB (MySQL compatible) | Bash | SQL triggers, stored procedure, views

Database Design

investigators

investigator_id (PK)
name, rank_title
department, phone, email

cases

case_id (PK)
case_title, case_type
status, registered_date
lead_investigator_id (FK)

evidence

evidence_id (PK)
case_id (FK), investigator_id (FK)
type, description, file_name
sha256_hash, collection_date
storage_location, status

chain_of_custody

custody_id (PK)
evidence_id (FK)
transferred_by, received_by (FK)
action, action_datetime
remarks

Relationships

One investigator leads many cases. One case holds many evidence items. One evidence item has many custody entries. Foreign keys reject any ID that does not exist.

Views: v_evidence_details and v_custody_history join the tables into readable reports.

Chain of Custody

1. Registered
Trigger logs a "Collected" entry automatically

2. Transferred
Procedure updates location + status and logs handover in one transaction

3. Reviewed
Full timeline of every handler, date and remark

Why the record can be trusted

- Tamper-proof: triggers block UPDATE and DELETE on custody records.
- Atomic: evidence status and log never disagree (transaction with rollback).
- Integrity: SHA-256 hash of each file proves it was not altered.

Features

Dashboard	Totals, counts by status, type and case, recent activity
Search	Keyword search across description, file, case, location
Filter	By evidence type, status or case ID
Register	New cases, evidence and investigators
Transfer	Hand over evidence and log custody
Reports	Case report with evidence list and custody trail

Demo Flow and Conclusion

Live demo steps

1. Add investigator (option 10)
2. Register case (option 5)
3. Register evidence (option 6)
4. Transfer evidence (option 7)
5. View custody trail (option 8)
6. Show dashboard (option 1)
7. Try DELETE on custody: blocked

Conclusion

DEMS replaces manual records with one accountable database.
Every handling step is recorded and cannot be altered.
Future scope: web dashboard, user login and roles, file upload with automatic hashing.

Thank you